DETECTING A PROPERTY OF THE FIBRES ON LOCALISATIONS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Property (B): a property of Noetherian rings that can be checked on local rings. assumed in the statement and used at proof line 12895, source lines 12820-12869. Justifies replacing the verification on fibres by a verification on their local rings.
- The fibre of a ring map over a prime, identified with the spectrum of the base change to the residue field. used at proof lines 12884-12890 with its description of the primes lying over the given one. Supplies the object whose localisation is computed.
- Identification of the localisation of a base changed algebra with the localisation at a chosen maximal ideal. used at proof lines 12890-12899. Is the computation that reduces conditions (1) and (3) to each other.

Source reference. Original lemma 0BK8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14551–14551. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00E6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 523–523. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $R \to \Lambda$ be a homomorphism of Noetherian rings. Assume P has property (B). The following are equivalent*

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(1) *the fibres of* $R \to \Lambda$ *have* P,
(2) *the fibres of* $R_{\mathfrak{p}} \to \Lambda_{\mathfrak{q}}$ *have* P *for all* $\mathfrak{q} \subset \Lambda$ *lying over* $\mathfrak{p} \subset R$, *and*
(3) *the fibres of* $R_{\mathfrak{m}} \to \Lambda_{\mathfrak{m}'}$ *have* P *for all maximal ideals* $\mathfrak{m}' \subset \Lambda$ *lying over* $\mathfrak{m}$ *in* R.

Proof. Let $\mathfrak{p} \subset R$ be a prime. Then the fibre over $\mathfrak{p}$ is the ring $\Lambda \otimes_R \kappa(\mathfrak{p})$ whose spectrum maps bijectively onto the subset of $\mathrm{Spec}(\Lambda)$ consisting of primes $\mathfrak{q}$ lying over $\mathfrak{p}$, see Algebra, Remark 00E6. For such a prime $\mathfrak{q}$ choose a maximal ideal $\mathfrak{q} \subset \mathfrak{m}'$ and set $\mathfrak{m} = R \cap \mathfrak{m}'$. Then $\mathfrak{p} \subset \mathfrak{m}$ and we have

$$(\Lambda \otimes_R \kappa(\mathfrak{p}))_{\mathfrak{q}} \cong (\Lambda_{\mathfrak{m}'} \otimes_{R_{\mathfrak{m}}} \kappa(\mathfrak{p}))_{\mathfrak{q}}$$

as $\kappa(\mathfrak{q})$-algebras. Thus (1), (2), and (3) are equivalent because by (B) we can check property P on local rings. $\square$